

2017-Ph-H1-Q2

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

A ball is projected upwards at an angle. Which row correctly describes the horizontal and vertical components of the ball's acceleration during the flight?

Key Concept:

- **Horizontal acceleration** in projectile motion = **0** (ignoring air resistance)
- **Vertical acceleration** = **constant downwards**, due to gravity $\rightarrow -9.8 \text{ m/s}^2$

 **Final Answer: C — Horizontal: 0, Vertical: constant**

Revision Tips:

- Projectiles always accelerate **vertically downward**.
- The horizontal component of velocity stays **constant**.
- The vertical component **changes** due to gravity.